
A2 BIOLOGY

6A: Microbiology

Bacteria

Characteristics of bacteria

- **Prokaryotic Cells:** Bacteria are single-celled organisms lacking a nucleus and membrane-bound organelles.
- **Size:** Typically, 0.5 to 5 micrometers in size.
- **Cell Structure:**
 - **Cell Wall:** Made of peptidoglycan, providing structural support.
 - **Plasma Membrane:** Phospholipid bilayer controlling entry and exit of substances.
 - **Cytoplasm:** Contains ribosomes (70S), DNA, and other molecules necessary for cellular processes.
 - **Nucleoid:** Region where the bacterial chromosome (circular DNA) is located.
 - **Plasmids:** Small, circular DNA molecules that can replicate independently of the bacterial chromosome and often carry genes for antibiotic resistance.
 - **Ribosomes:** Sites of protein synthesis, smaller than those in eukaryotes.
 - **Flagella:** Tail-like structures that enable motility in some bacteria.
 - **Pili:** Hair-like structures that allow attachment to surface and conjugation (transfer of genetic material).

Reproduction

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- **Binary Fission:** Primary method of reproduction, where a bacterial cell divides into two genetically identical daughter cells.
 - **Generation Time:** Time taken for a bacterial population to double, varying widely among species.
 - **Mutation and Genetic Variation:** Although binary fission produces identical clones, mutations can occur, leading to genetic variation. Additionally, bacteria can acquire new genetic material through:
 - **Conjugation:** Transfer of DNA between two bacterial cells via a pilus.
 - **Transformation:** Uptake of free DNA from the environment.³
 - **Transduction:** Transfer of bacterial DNA by a virus (bacteriophage).

Classification

- **Gram Staining:**
 - **Gram-Positive Bacteria:** Stain purple due to thick peptidoglycan layer.
 - **Gram-Negative Bacteria:** Stain pink/red due to thin peptidoglycan layer and an outer lipid membrane.
- **Oxygen Requirements:**
 - **Obligate Aerobes:** Require oxygen for respiration.
 - **Obligate Anaerobes:** Cannot survive in the presence of oxygen.
 - **Facultative Anaerobes:** Can survive with or without oxygen.
- **Shape:**
 - **Cocci:** Spherical (e.g., Streptococcus).
 - **Bacilli:** Rod-shaped (e.g., Escherichia coli).
 - **Spirilla:** Spiral-shaped (e.g., Spirillum).

Roles

- **Pathogenic Bacteria:** Cause diseases by producing toxins or damaging host tissues. Example: Mycobacterium tuberculosis (causes tuberculosis).
- **Beneficial Bacteria:**
 - **Nitrogen Fixation:** Certain bacteria (e.g., Rhizobium) convert atmospheric nitrogen into a form usable by plants.
 - **Decomposition:** Decompose dead organic matter, recycling nutrients in ecosystems.

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- **Symbiotic Relationships:** Gut bacteria aid in digestion and production of vitamins.

Viruses

Characteristics

- **Acellular Entities:** Viruses are not considered living organisms as they lack cellular structure and cannot carry out metabolic processes independently.
- **Structure:**
 - **Genetic Material:** DNA or RNA, which can be single-stranded or double-stranded.
 - **Capsid:** Protein coat that encloses the genetic material, made up of capsomeres.
 - **Envelope:** Some viruses have a lipid envelope derived from the host cell membrane, with glycoproteins that assist in host cell recognition and entry.
 - **Shapes:** Vary from helical to icosahedral, and complex structures.

Replication

- **Attachment:** Virus binds to specific receptors on the host cell surface.
- **Penetration:** The viral genome enters the host cell, either by direct injection (bacteriophages) or by endocytosis (animal viruses).
- **Replication and Transcription:** The viral genome is replicated using the host's machinery. DNA viruses typically replicate in the nucleus, while RNA viruses often replicate in the cytoplasm.
- **Assembly:** New viral particles (virions) are assembled from the replicated genome and proteins.
- **Release:** New virions are released either by lysis of the host cell or by budding (enveloped viruses), where the virus acquires its lipid envelope.

Types

- **DNA Viruses:** Replicate their DNA in the host cell's nucleus. Example: Herpesviruses.

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- **RNA Viruses:** Replicate in the host cell's cytoplasm. Example: Influenza virus.
 - **Retroviruses:** RNA viruses that reverse transcribe their RNA into DNA, integrating into the host genome. Example: HIV.

Pathogenicity

- **Lytic Cycle:** The virus hijacks the host cell's machinery to produce new virions, leading to cell lysis and death. Example: T4 bacteriophage.
- **Lysogenic Cycle:** The viral DNA integrates into the host genome and remains dormant until triggered to enter the lytic cycle. Example: Lambda phage.
- **Host Specificity:** Viruses are often host-specific and may even target specific cell types within a host, known as tissue tropism.

Impact

- **Human Diseases:** Viruses cause a range of diseases, from the common cold (rhinoviruses) to more serious conditions like AIDS (HIV), hepatitis (Hepatitis B and C), and COVID-19 (SARS-CoV-2).
- **Economic Impact:** Viral diseases in plants and animals can lead to significant agricultural losses. Example: Tobacco Mosaic Virus in plants.
- **Medical Applications:**
 - **Vaccines:** Used to prevent viral infections by inducing immunity. Example: MMR vaccine (measles, mumps, rubella).
 - **Gene Therapy:** Modified viruses (e.g., adenoviruses) are used as vectors to deliver therapeutic genes into human cells.

Bacterial colonies

- **Bacterial Growth:** Refers to an increase in the number of bacterial cells rather than an increase in size. Bacteria reproduce by binary fission, where one cell divides into two identical daughter cells.
- **Population Doubling:** The time it takes for the bacterial population to double is known as the **generation time**. This can vary depending on the species and environmental conditions.

Bacterial Growth Curve

Bacterial growth in a closed system (batch culture) typically follows four distinct phases:

- **Lag Phase:**
 - **Adaptation:** Bacteria adapt to their new environment, synthesizing enzymes and other molecules necessary for growth.
 - **Slow Growth:** Little or no cell division occurs, but metabolic activity is high.
 - **Preparation:** Cells prepare for DNA replication and division.
- **Log (Exponential) Phase:**
 - **Rapid Growth:** Bacteria undergo rapid cell division, and the population doubles at a constant rate.
 - **Optimal Conditions:** Growth is at its maximum rate, and cells are most susceptible to antibiotics.
 - **Exponential Increase:** The population size increases logarithmically.
- **Stationary Phase:**
 - **Growth Rate Equals Death Rate:** The number of new cells produced equals the number of cells dying, resulting in a stable population size.
 - **Nutrient Depletion:** Essential nutrients become depleted, and waste products accumulate.
 - **Limited Resources:** The growth rate slows down due to limited availability of resources.

- **Death (Decline) Phase:**

- **Cell Death Exceeds Growth:** The death rate exceeds the rate of new cell formation, leading to a decline in the population size.
- **Toxic Byproducts:** Accumulation of toxic byproducts and complete depletion of nutrients cause cells to die.

Factors Affecting Bacterial Growth:

- **Temperature:**

- **Optimum Temperature:** Bacteria have an optimum temperature range for growth. Extremes in temperature can inhibit growth or kill bacteria.
- **Psychrophiles:** Thrive at low temperatures (0-20°C).
- **Mesophiles:** Thrive at moderate temperatures (20-45°C).
- **Thermophiles:** Thrive at high temperatures (45-80°C).

- **pH:**

- **Optimum pH:** Bacteria have an optimum pH range, typically around neutral (pH 6.5-7.5) for most bacteria.
- **Acidophiles:** Grow in acidic environments (pH < 6).
- **Alkaliphiles:** Grow in alkaline environments (pH > 8).

- **Oxygen Concentration:**

- **Obligate Aerobes:** Require oxygen for growth.
- **Obligate Anaerobes:** Cannot tolerate oxygen and grow only in its absence.
- **Facultative Anaerobes:** Can grow with or without oxygen.
- **Microaerophiles:** Require low levels of oxygen.
- **Aerotolerant Anaerobes:** Do not use oxygen but are not harmed by its presence.

- **Nutrient Availability:**

- **Carbon Source:** Bacteria require a carbon source for energy and cellular structures. Autotrophs use CO₂, while heterotrophs use organic compounds.
- **Nitrogen Source:** Essential for protein and nucleic acid synthesis. Bacteria may use ammonia, nitrate, or nitrogen gas (via nitrogen fixation).
- **Growth Factors:** Some bacteria require specific vitamins, amino acids, or other organic molecules that they cannot synthesize themselves.

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- **Water Availability:**
 - **Osmotic Pressure:** Bacteria require an appropriate balance of water in their environment. High salt or sugar concentrations can inhibit growth by causing osmotic stress.
 - **Light:**
 - **Phototrophic Bacteria:** Use light as an energy source. However, most pathogenic bacteria are not phototrophic and grow independently of light

Measuring Bacterial Growth

Direct Methods:

- **Viable Count (Colony-Forming Units - CFUs):**
 - **Spread Plate and Pour Plate Methods:** Serial dilutions of a bacterial culture are plated on agar. After incubation, colonies are counted, and the number of viable bacteria in the original sample is calculated.
 - **Advantages:** Provides an accurate count of live bacteria.
 - **Disadvantages:** Time-consuming and only counts bacteria that can grow under the conditions provided.
- **Microscopic Count:**
 - **Counting Chamber (e.g., Hemocytometer):** A known volume of culture is placed on a special slide, and cells are counted under a microscope.
 - **Advantages:** Quick and easy.
 - **Disadvantages:** Does not differentiate between live and dead cells, and small or motile cells may be difficult to count accurately.

Indirect Methods:

- **Turbidity (Optical Density - OD):**
 - **Spectrophotometry:** The turbidity of a bacterial culture, measured by the absorbance of light at a specific wavelength, correlates with cell density.
 - **Advantages:** Rapid and non-destructive.

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- **Disadvantages:** Does not distinguish between live and dead cells and requires calibration for accurate cell counts.
 - **Dry Weight:**
 - **Measurement:** Bacterial cells are filtered, dried, and weighed to estimate biomass.
 - **Advantages:** Accurate for large-scale cultures.
 - **Disadvantages:** Time-consuming and not suitable for small cultures.
 - **Biochemical Activity:**
 - **Measurement of Metabolic Products:** The rate of production of CO₂, acid, or other metabolites can be used to estimate bacterial growth.
 - **Advantages:** Useful for monitoring specific metabolic activities.
 - **Disadvantages:** Indirect and may not correlate directly with cell numbers.

Culturing bacteria

Types of media

- **Nutrient Broth/Agar:**
 - **General Purpose:** Supports the growth of a wide range of bacteria by providing basic nutrients like peptones, amino acids, salts, and vitamins.
 - **Examples:** Nutrient agar, Luria-Bertani (LB) broth.
- **Selective Media:**
 - **Purpose:** Contains inhibitors that suppress the growth of unwanted bacteria, allowing only specific types to grow.
 - **Examples:** MacConkey agar (selects for Gram-negative bacteria), Mannitol salt agar (selects for Staphylococcus).
- **Differential Media:**
 - **Purpose:** Contains indicators that reveal differences between bacterial species based on their metabolic activity.

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- **Examples:** Blood agar (differentiates bacteria based on hemolysis), MacConkey agar (differentiates based on lactose fermentation).
 - **Enrichment Media:**
 - **Purpose:** Encourages the growth of specific bacteria while inhibiting others, often used when the desired bacteria are in low numbers.
 - **Examples:** Selenite broth (used to enrich Salmonella).

Aseptic Techniques

- **Purpose:** Prevents contamination of cultures and the environment. Includes practices such as sterilizing equipment, using flame sterilization, and working near a Bunsen burner flame.
- **Sterilization Methods:**
 - **Autoclaving:** Uses steam under pressure to sterilize media, equipment, and waste.
 - **Filtration:** Sterilizes heat-sensitive liquids by removing bacteria and other microorganisms.
 - **UV Radiation:** Kills bacteria on surfaces and in the air.

Colony morphology

- **Observation:** Bacterial colonies on agar plates can be distinguished by their shape, size, color, texture, and margin.
- **Types of Colonies:**
 - **Circular:** Round colonies.
 - **Irregular:** Colonies with uneven edges.
 - **Filamentous:** Colonies with a thread-like appearance.
 - **Mucoid:** Slimy colonies often indicative of capsule production.

Importance of Bacterial Colony Growth in Various Fields

Medical Microbiology:

- **Diagnosis of Infections:** Culturing bacteria from clinical samples helps identify the causative agents of infections.
- **Antibiotic Sensitivity Testing:** Bacterial colonies are tested against antibiotics to determine the most effective treatment.

Food Microbiology:

- **Food Safety:** Monitoring bacterial growth in food products helps prevent foodborne illnesses.
- **Fermentation:** Controlled growth of specific bacteria is essential in the production of fermented foods like yogurt, cheese, and sauerkraut.

Environmental Microbiology:

- **Bioremediation:** Bacteria are used to degrade pollutants and clean up contaminated environments.
- **Wastewater Treatment:** Bacterial colonies break down organic matter in sewage, making it safer for disposal or reuse.

Industrial Microbiology:

- **Production of Antibiotics:** Bacterial colonies are used in the large-scale production of antibiotics, enzymes, and other industrially important products.
- **Bioengineering:** Genetic engineering of bacteria for the production of recombinant proteins and other biotechnological applications.

How Bacteria and Viruses Invade the Body

In microbiology, understanding how bacteria and viruses invade the human body is key to grasping how infections occur. Bacteria and viruses have distinct methods of entry, replication, and survival, causing disease in different ways.

Bacterial Invasion

Bacteria are prokaryotic microorganisms that can cause disease by invading the body, reproducing, and producing toxins. Their invasion involves multiple stages:

1. Entry into the Body

Bacteria can enter the body through various routes, including:

Inhalation: Bacteria can enter the respiratory system through inhalation of airborne droplets, such as in the case of *Mycobacterium tuberculosis* (causing tuberculosis).

Ingestion: Bacteria can enter the gastrointestinal tract through the consumption of contaminated food or water, as seen with *Salmonella* or *Escherichia coli*.

Breaks in the Skin: Bacteria can enter through wounds, cuts, or surgical incisions. *Staphylococcus aureus* is an example of a bacterium that can cause skin infections after penetrating the skin barrier.

Mucosal Surfaces: Bacteria can invade through mucous membranes, such as those in the eyes, mouth, or genitals. For instance, *Neisseria gonorrhoeae* causes gonorrhea by infecting the mucosal membranes.

2. Adhesion and Colonization

Once inside the body, bacteria must adhere to host cells to colonize and establish an infection. They use adhesins, which are surface proteins that bind to receptors on the host cell. For example, *E. coli* uses fimbriae (hair-like projections) to attach to cells in the urinary tract, causing urinary tract infections.

After adhesion, bacteria begin to colonize the area, multiplying rapidly. Some bacteria form biofilms, which are protective layers that help them resist the host immune response and antibiotics.

3. Evasion of the Immune System

Bacteria have evolved several mechanisms to evade the host's immune defenses, including:

1. **Capsules:** Many bacteria have a capsule (polysaccharide layer) that prevents phagocytosis by immune cells, as seen in *Streptococcus pneumoniae*.
2. **Toxins:** Some bacteria produce toxins that can damage or kill host cells. For instance, *Clostridium botulinum* produces botulinum toxin, which can cause paralysis by interfering with nerve function.
3. **Antigenic Variation:** Some bacteria, like *Neisseria gonorrhoeae*, can change their surface antigens to avoid detection by the immune system.

4. Damage to Host Tissues

Bacterial infections often lead to tissue damage through:

Exotoxins: These are toxins secreted by bacteria that can directly harm host cells. *Clostridium tetani* produces tetanus toxin, which affects the nervous system.

Endotoxins: These are toxins released when gram-negative bacteria (like *E. coli*) die and their cell walls break down. Endotoxins can trigger severe inflammation and fever.

Enzymes: Some bacteria produce enzymes that break down tissues and allow the bacteria to spread more easily. *Streptococcus pyogenes* produces enzymes that degrade connective tissue, causing diseases like necrotizing fasciitis.

Viral Invasion

Viruses are acellular pathogens that must enter host cells to replicate, as they rely on the host's cellular machinery. Viral invasion follows a distinct pathway:

1. Entry into the Body

Viruses enter the body through similar routes as bacteria but must specifically target host cells for replication. Common routes include:

Inhalation: Respiratory viruses like influenza or the coronavirus (SARS-CoV-2) enter through the respiratory tract.

Ingestion: Viruses like rotavirus infect the gastrointestinal tract after being ingested.

Mucosal Surfaces and Skin: Some viruses, such as the human papillomavirus (HPV), enter through the skin or mucosal membranes, often through micro-abrasions.

Bloodstream: Some viruses, like HIV and hepatitis B, can be transmitted through blood-to-blood contact, such as through needle sharing or transfusions.

2. Attachment to Host Cells

Viruses must first bind to specific receptors on the surface of the host cell to initiate infection. The viral attachment proteins (often glycoproteins) on the virus's outer surface bind to complementary receptors on the host cell membrane.

For example:

- HIV binds to the CD4 receptor on helper T cells using its glycoprotein gp120.
- Influenza viruses bind to sialic acid receptors on respiratory epithelial cells using hemagglutinin proteins.

This specificity determines which cells and tissues a virus can infect.

3. Entry into Host Cells

After attachment, the virus enters the host cell through various mechanisms:

Endocytosis: The host cell membrane engulfs the virus, bringing it inside as a vesicle. This is typical of many viruses, including the influenza virus.

Membrane Fusion: Enveloped viruses, like HIV, fuse their outer membrane with the host cell membrane, allowing their genetic material to enter directly into the cytoplasm.

4. Replication

Once inside, the virus hijacks the host cell's machinery to replicate its genetic material and produce viral proteins. Depending on the virus, the replication process varies:

DNA Viruses: Replicate their DNA in the host cell's nucleus (e.g., herpesviruses).

RNA Viruses: Replicate their RNA directly in the cytoplasm (e.g., the rabies virus).

Retroviruses like HIV reverse transcribe their RNA into DNA using the enzyme reverse transcriptase, which is then integrated into the host cell's genome.

5. Evasion of the Immune System

Viruses have evolved several strategies to evade detection by the host's immune system:

Latency: Some viruses, like herpesviruses, can remain dormant (latent) in host cells, avoiding immune detection for long periods.

Inhibition of Immune Responses: Viruses like HIV destroy or inhibit immune cells (T cells), weakening the immune response.

Antigenic Drift and Shift: Influenza viruses can undergo small (drift) or major (shift) changes in their surface proteins, helping them evade immunity built from previous infections or vaccines.

6. Release from Host Cells

After replication, viruses must exit the host cell to infect new cells. This can occur through:

Lysis: Some viruses, like adenoviruses, cause the host cell to burst (lyse), releasing new viral particles.

Budding: Enveloped viruses, like HIV or influenza, bud off from the host cell membrane, acquiring a portion of the membrane as their envelope.

7. Damage to Host Cells and Tissues

Viral infections often result in host cell damage and contribute to disease in several ways:

Direct Cytopathic Effects: Viruses can directly kill or damage host cells as part of their replication cycle, as seen in viral-induced apoptosis (cell death).

Immune-Mediated Damage: In some cases, the immune system's response to the virus can cause tissue damage. For instance, in hepatitis, the immune system attacks virus-infected liver cells, leading to liver damage.

Human Defense Mechanisms

To counter bacterial and viral infections, the body relies on several lines of defense:

Physical Barriers: The skin, mucous membranes, and stomach acid act as the first line of defense.

Innate Immune Response: Includes phagocytic cells (macrophages, neutrophils), inflammation, and the release of antiviral proteins like interferons.

Adaptive Immune Response: Involves the activation of T cells and B cells, which target specific pathogens. B cells produce antibodies that neutralize viruses, while cytotoxic T cells kill virus-infected cells.

CASE STUDIES: *Mycobacterium tuberculosis* (M. tuberculosis)

Mycobacterium tuberculosis is the bacterium responsible for **tuberculosis (TB)**, an infectious disease primarily affecting the lungs. Here are the key points for your exam:

Characteristics

- **Rod-shaped, non-motile, aerobic** bacterium.
- **Waxy cell wall** (mycolic acid) makes it **acid-fast** and resistant to many treatments.
- Identified by **acid-fast staining** (e.g., Ziehl-Neelsen stain).
- **Slow growth**, with a generation time of 15-20 hours.

Transmission and Pathogenesis

- Transmitted via **airborne droplets** when infected individuals cough or sneeze.
- **Primary infection**: Bacteria reach the **alveoli**, where they are engulfed by **macrophages**.
- **Latent TB**: Bacteria can remain dormant in **granulomas**. In this state, individuals are non-contagious but can develop active TB later.
- **Active TB**: Granulomas break down, bacteria spread, and the person becomes symptomatic and contagious.

Symptoms

- **Pulmonary TB**: Persistent **cough**, often with **blood**, **fever**, **night sweats**, **weight loss**, and **chest pain**.
- **Extrapulmonary TB**: Can affect organs like the kidneys and brain.

Diagnosis

- **Tuberculin Skin Test (TST)** or **Mantoux test**.
- **Chest X-rays** to identify lung damage.
- **Sputum test** for bacterial detection.

Treatment

- First-line antibiotics: **Isoniazid, Rifampicin, Pyrazinamide, Ethambutol** (typically for 6 months).
- Long treatment courses are necessary to prevent relapse and resistance.
- **Multidrug-resistant TB (MDR-TB)** and **Extensively drug-resistant TB (XDR-TB)** are major concerns.

Prevention

- **BCG vaccine** provides protection, especially for children.
- **Infection control** measures in healthcare and crowded settings.

CASE STUDY: HIV (Human Immunodeficiency Virus)

HIV is the virus that causes **AIDS** (Acquired Immunodeficiency Syndrome). It weakens the immune system, making the body vulnerable to infections. Here's what you need to know:

What is HIV?

- **Retrovirus** that attacks **CD4+ T cells** (important immune cells).
- HIV carries **RNA** and uses an enzyme called **reverse transcriptase** to turn its RNA into DNA, which it inserts into the host's cells.

How is HIV Spread?

- **Sexual contact** (unprotected sex with an infected person).
- **Sharing needles** (e.g., drug use).
- **Mother-to-child** transmission during childbirth or breastfeeding.
- Contact with infected **blood, semen, or vaginal fluids**.

How Does HIV Affect the Body?

- **Early Stage (Acute HIV):** Within a few weeks of infection, some people experience flu-like symptoms (fever, sore throat, rash).
- **Chronic Stage (Latent HIV):** After the initial stage, HIV becomes less active but still present. People may not feel sick for years.
- **AIDS:** HIV destroys so many CD4+ cells that the immune system can't protect against infections and diseases, leading to AIDS.

Symptoms

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- **Early HIV:** Flu-like symptoms (fever, sore throat, rash).
 - **Later Stages:** Weight loss, night sweats, infections like TB or pneumonia.

How is HIV Diagnosed?

- **HIV Test:** Blood tests that look for HIV antibodies or the virus itself.
- **CD4 Count:** Measures how many CD4 cells you have; a low count means the immune system is weakened.

Treatment

- **Antiretroviral Therapy (ART):** A combination of medicines that help control the virus.
 - ART can't cure HIV, but it helps people live longer, healthier lives.
- **Lifelong Treatment:** HIV-positive people need to stay on ART for life to manage the virus.

How to Prevent HIV?

- **Do not share needles.**
- **PrEP:** Medication for people at high risk of HIV to reduce their chances of getting infected.
- HIV-positive mothers can take medicine to prevent passing the virus to their babies.

6B: Immunity

Non-Specific Responses to Infection

The non-specific immune response is the body's first line of defense against pathogens and works the same way for all infections. It includes:

1. Physical Barriers

- Skin: Acts as a barrier to prevent pathogens from entering.
- Mucous membranes: Line the respiratory, digestive, and other tracts, trapping pathogens.
- Cilia: Tiny hair-like structures that sweep mucus (and trapped pathogens) out of the airways.

2. Inflammation

- Inflammatory response: When tissues are damaged or infected, they release chemicals (like histamines) causing inflammation. This leads to:
 - Increased blood flow to the area.
 - Swelling, redness, heat, and pain.
 - Tissue fluids dilute pathogens
 - White blood cells, like phagocytes, move to the site of infection.

3. Phagocytosis

- Phagocytes (e.g., macrophages, neutrophils) engulf and digest pathogens through a process called phagocytosis.

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- The phagocyte surrounds and engulfs the pathogen, then breaks it down using enzymes.

4. Fever

- Fever: The body raises its temperature to slow down pathogen replication and enhance the immune response.

Specific Responses to Infection

The specific immune response (adaptive immunity) targets specific pathogens and involves lymphocytes. It has two main components:

1. B Lymphocytes (Humoral Response)

- B cells produce antibodies that are specific to antigens on pathogens.
 - Antibodies bind to antigens, marking the pathogen for destruction.
 - Some B cells become memory cells, which remember the pathogen for faster responses in future infections.

2. T Lymphocytes (Cell-Mediated Response)

- Helper T cells: Activate other immune cells, like B cells and cytotoxic T cells.
- Cytotoxic T cells: Directly destroy infected cells by releasing chemicals.
- Memory T cells: Remember the pathogen and help mount a quicker response if the infection recurs.

Key Features

- Specificity: Each response targets a specific pathogen.
- Memory: Memory cells ensure faster, stronger responses upon reinfection.

Step-by-Step Process of the Specific Immune Response

The specific immune response is a targeted defense mechanism against pathogens and involves the activation of B cells and T cells. Here's the process:

1. Recognition of Pathogen (Antigen Presentation)

- Pathogen enters the body and is engulfed by a phagocyte (e.g., macrophage).
- The phagocyte ingests the pathogen and presents its antigens on its surface using MHC (Major Histocompatibility Complex) proteins.
- The phagocyte becomes an antigen-presenting cell (APC) and moves to a lymph node.

2. Activation of Helper T Cells

- APCs present the pathogen's antigens to Helper T cells (CD4+ T cells) in the lymph node.
- If a Helper T cell with a receptor specific to the antigen is found, it binds to the APC and becomes activated.
- Activated Helper T cells begin to multiply (clonal expansion) and release cytokines (chemical signals).

3. Activation of B Cells (Humoral Response)

- B cells encounter the pathogen and bind to its antigen with their specific antibodies on their surface.
- However, B cells also need activation from Helper T cells. Activated Helper T cells recognize the antigen displayed on the B cell and bind to it.
- Cytokines from the Helper T cell fully activate the B cell.

4. Clonal Expansion and Differentiation of B Cells

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- The activated B cell divides to produce many plasma cells and memory B cells.
 - Plasma cells produce large amounts of antibodies specific to the pathogen's antigen. These antibodies neutralize the pathogen or mark it for destruction by other immune cells.
 - Memory B cells are long-lived and provide faster antibody production if the pathogen invades again in the future.

5. Activation of Cytotoxic T Cells (Cell-Mediated Response)

- Cytotoxic T cells (CD8+ T cells) are also activated by Helper T cells.
- Once activated, Cytotoxic T cells recognize infected cells (cells displaying the pathogen's antigens via MHC I) and bind to them.
- They release perforins and granzymes to destroy the infected cells.

6. Elimination of Pathogen

- Antibodies from plasma cells neutralize pathogens in body fluids.
- Cytotoxic T cells kill infected cells directly, stopping the spread of infection.

7. Formation of Memory Cells

- After the infection is cleared, some B cells and T cells become memory cells.
- Memory B cells and Memory T cells remain in the body, allowing for a faster and stronger immune response if the same pathogen infects again in the future.

This process ensures that the immune system specifically targets the pathogen and prepares for quicker responses if reinfection occurs.

Developing Immunity

Immunity refers to the body's ability to resist infections. There are two main types: **active immunity** and **passive immunity**, each of which can be **natural** or **artificial**.

1. Active Immunity

- **Definition:** Immunity that develops when the body produces its own antibodies and memory cells in response to an infection or a vaccine.
- **Natural Active Immunity:**
 - Occurs when you are exposed to a pathogen (e.g., getting infected by a virus), and your immune system generates a response.
 - Results in long-lasting immunity due to the formation of **memory cells**.
- **Artificial Active Immunity:**
 - Achieved through **vaccination**, where a person is exposed to a weakened or inactive form of the pathogen.
 - This stimulates the immune system to produce **antibodies** and memory cells without causing disease.

2. Passive Immunity

- **Definition:** Immunity acquired through the transfer of antibodies from another source, without the immune system producing them.
- **Natural Passive Immunity:**
 - Happens when a baby receives **antibodies from the mother** through the placenta during pregnancy or through **breast milk**.
 - Provides immediate protection but is **short-lived** because no memory cells are formed.
- **Artificial Passive Immunity:**
 - Involves receiving **antibodies** from an external source, such as an **injection of antiserum** (e.g., for snake bites or certain infections).
 - Offers immediate protection but does not lead to long-term immunity.

3. Vaccination

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- **Vaccines** contain antigens from dead or weakened pathogens. They trigger the immune response, leading to the production of **antibodies** and **memory cells** without causing illness.
 - **Booster shots** may be needed to ensure long-term immunity by stimulating memory cells.

4. Herd Immunity

- Occurs when a large portion of a population becomes immune to a disease, either through vaccination or previous infection.
- This reduces the spread of disease and protects those who cannot be vaccinated (e.g., babies, immunocompromised individuals).

Antibiotics And Resistance

Antibiotics are drugs used to treat bacterial infections by either killing bacteria or stopping their growth. However, misuse and overuse of antibiotics have led to **antibiotic resistance**, where bacteria evolve to survive the effects of these drugs.

1. How Antibiotics Work

- **Bactericidal antibiotics:** Kill bacteria by disrupting vital processes like cell wall synthesis (e.g., penicillin).
- **Bacteriostatic antibiotics:** Prevent bacterial growth and reproduction (e.g., tetracyclines).

Antibiotics do **not** work against viral infections like colds or flu, as viruses lack the structures antibiotics target.

2. Antibiotic Resistance

- **Definition:** When bacteria develop the ability to survive exposure to antibiotics that would normally kill them.
- **Mechanisms of Resistance:**
 - **Mutations:** Random changes in bacterial DNA that make antibiotics ineffective.
 - **Horizontal gene transfer:** Bacteria can share resistance genes through processes like **conjugation** (bacterial "mating").

3. Causes of Antibiotic Resistance

- **Overuse:** Taking antibiotics when they aren't necessary, such as for viral infections.
- **Incomplete courses:** Not finishing the prescribed antibiotic course, allowing some bacteria to survive and develop resistance.
- **Use in agriculture:** Excessive use of antibiotics in livestock promotes the development of resistant bacteria.

4. Consequences of Resistance

- **Superbugs:** Strains of bacteria that are resistant to multiple antibiotics, making infections difficult or impossible to treat (e.g., **MRSA**).
- Resistant infections result in longer illness, increased medical costs, and higher mortality rates.

5. Combating Antibiotic Resistance

- **Appropriate use of antibiotics:** Only taking antibiotics when prescribed and completing the full course.
- **Development of new antibiotics:** Pharmaceutical research focuses on discovering new drugs to target resistant bacteria.
- **Alternatives to antibiotics:** Strategies like phage therapy (using viruses to kill bacteria) and vaccinations to prevent bacterial infections.
- **Antibiotic stewardship:** Programs that promote the responsible use of antibiotics in healthcare and agriculture.

6C: Decomposition And Forensics

Microorganisms And Decomposition

1. Role in Decomposition:

Microorganisms such as bacteria and fungi are crucial in breaking down dead organic matter (plants, animals, etc.). They use the complex organic molecules as an energy source and convert them into simpler substances.

Decomposition Process:

Microbes secrete enzymes that break down large molecules (proteins, fats, carbohydrates) into smaller, soluble ones.

These simpler compounds are absorbed by the microorganisms for their own metabolism.

As a result, the dead matter is broken down into nutrients and gasses.

2. Recycling of Carbon:

During decomposition, microorganisms play an essential role in the **carbon cycle**:

- **Respiration of Microbes:** As microbes break down organic material, they release carbon dioxide through respiration. This CO₂ re-enters the atmosphere, where it is used by plants for photosynthesis.
- **Release of Organic Matter into the Soil:** Some of the carbon from decomposed organisms is incorporated into the soil, enriching it with nutrients like humus, which supports plant growth.

The decomposition process ensures the continuous cycling of carbon between the biosphere, atmosphere, and geosphere, maintaining ecological balance. This recycling is a key aspect of how ecosystems sustain life.

Forensic Science & The Time Of Death

Key Concepts for Determining Time of Death:

Extent of Decomposition:

Decomposition stages, as discussed earlier, can give clues about how long the body has been dead based on physical changes in the body such as discoloration, bloating, and liquefaction.

Stage of Succession:

This refers to the different stages of insect colonization of a dead body. Different species of insects, such as blowflies and beetles, arrive at different stages of decomposition, helping to estimate the time since death.

Forensic Entomology:

The study of insects, particularly flies and beetles, found on or around a body. The development of insects like maggots (fly larvae) can provide a timeline for how long the body has been exposed.

Body Temperature (Algor Mortis):

The body's temperature decreases after death as it cools down to match the surrounding environment. This cooling can be measured to estimate the time of death, especially in the first 24-48 hours.

Change in body temperature can be affected by clothing, fat levels, mode of death and the surrounding environment.

Degree of Muscle Contraction (Rigor Mortis):

After death, the body goes through rigor mortis, where the muscles stiffen due to chemical changes. This starts within a few hours of death, peaks at around 12 hours, and subsides after 24-36 hours. The timing of these changes helps estimate the time of death.

Investigating Time of Death:

Determining the exact time of death can be challenging, particularly after 48 hours.

Stages of Decay

After death, the body undergoes a series of decay processes. These processes are driven primarily by the action of microorganisms (bacteria and fungi) and insects. Decay is divided into several stages:

1. Fresh Stage (Autolysis):

Immediately after death, cells and tissues break down due to enzymatic activity in a process known as autolysis.

No visible external signs of decay appear during this stage.

Oxygen levels in the body decrease rapidly, creating an environment suitable for anaerobic microorganisms to start multiplying.

2. Bloating Stage:

This stage is characterized by the production of gases such as methane, ammonia, hydrogen sulfide, and carbon dioxide, produced by bacteria.

These gases cause the body to bloat and expand.

Microbial activity increases, especially by anaerobic bacteria that flourish in the oxygen-deprived tissues.

This is the stage where forensic entomology becomes important. Flies lay eggs on the body, which hatch into maggots that begin feeding on soft tissues.

3. Active Decay (Putrefaction):

In this stage, the body's soft tissues begin to break down rapidly.

The skin starts to break down, and tissues liquefy due to the activity of bacteria and maggots. Foul-smelling gases are released during this process.

The maggots feed on tissues, accelerating decomposition.

Beetles and other insects start arriving to lay their eggs, which feed on decaying tissue or on other larvae like maggots.

4. Advanced Decay:

By this stage, much of the body's soft tissues have decomposed.

Insects and microorganisms continue to consume remaining tissues.

The body starts to dry out as the supply of soft tissues diminishes, leading to a reduction in microbial activity.

In this stage, forensic experts analyze insect colonization patterns to determine the post-mortem interval.

5. Dry/Skeletal Stage:

In the final stage, the body consists mostly of bones, hair, and any other resistant tissues.

The rate of decomposition slows, as microorganisms and insects have little organic material left to consume.

At this point, the body has been reduced to its skeleton and other durable remnants.

Role of Microorganisms and Insects:

- **Microorganisms** (bacteria and fungi) play a key role in breaking down organic material. These decomposers recycle nutrients back into the ecosystem.
- **Insects** (particularly flies, beetles, and their larvae) contribute by consuming soft tissues, aiding in the breakdown of the body.

Forensic Importance of Microorganisms:

The stages of decay, especially the presence of specific insects and the level of decomposition, are critical clues in forensic science for estimating the **post-mortem interval (PMI)**—the time since death. Forensic experts rely on:

- **Insect life cycles** (especially fly larvae development).
- The presence of specific species that appear at certain stages of decay.
- The extent of tissue breakdown and body condition.

These stages help forensic scientists piece together a timeline of events, which is essential in legal and criminal investigations.

Polymerase Chain Reaction (PCR) in Forensics

PCR (Polymerase Chain Reaction) is a crucial technique in forensic science used for amplifying small quantities of DNA to generate enough material for analysis. This method is particularly useful when DNA samples are degraded or available only in trace amounts, such as in crime scene investigations.

1. What is PCR?

PCR is a **molecular biology technique** used to create millions or even billions of copies of a specific DNA sequence in a short amount of time. It involves repeated cycles of DNA denaturation, annealing of primers, and elongation using a DNA polymerase enzyme.

2. Key Steps in PCR:

The process involves three main stages, which are repeated in multiple cycles to amplify the DNA:

- **Denaturation (95°C):**
 - The DNA sample is heated, causing the two strands of the double helix to separate, breaking the hydrogen bonds between the bases.
- **Annealing (50–65°C):**
 - Short DNA sequences called **primers**, designed to match the target DNA region, bind to the separated strands. The primers mark the starting point for DNA replication.
- **Extension (72°C):**
 - **Taq polymerase**, a heat-stable enzyme, extends the primers by adding complementary nucleotides to the DNA strands, synthesizing a new strand of DNA.

This cycle is repeated typically 25-40 times, resulting in the exponential amplification of the target DNA sequence.

3. Applications of PCR in Forensics:

- **DNA Profiling:**

- PCR is often used in DNA profiling to analyze genetic markers in a forensic sample. Specific regions of the DNA, such as short tandem repeats (STRs), are amplified and compared across samples to determine identity or familial relationships.
- PCR allows forensic scientists to develop DNA profiles even from very small or degraded samples like hair, blood, or skin cells left at a crime scene.

- **Analysis of Degraded DNA:**

- PCR is invaluable when dealing with degraded or old DNA, such as samples from bones or ancient remains, as it can amplify the DNA regions that are still intact, even if most of the DNA has broken down.

- **Mitochondrial DNA Analysis:**

- For cases where nuclear DNA is unavailable (e.g., when analyzing hair shafts or ancient remains), mitochondrial DNA (mtDNA) can be amplified using PCR. MtDNA is inherited maternally and is more abundant in cells than nuclear DNA, making it useful for certain forensic investigations.

- **Crime Scene Investigation:**

- PCR helps in analyzing trace amounts of biological material found at crime scenes, including blood, semen, saliva, skin cells, and even fingerprints. This is essential for linking a suspect to a crime scene or victim.

- **Identifying Victims in Disasters:**

- PCR is used in **mass disaster identification** (e.g., plane crashes, natural disasters) where DNA samples are degraded due to environmental conditions. The technique allows for the rapid identification of victims by amplifying DNA for comparison with family members' DNA profiles.

- **Exoneration and Legal Proceedings:**

- PCR has played a pivotal role in **exonerating innocent individuals**. By analyzing DNA evidence from past cases, PCR can produce reliable DNA profiles that may overturn wrongful convictions.

4. Advantages of PCR in Forensics:

- **Sensitivity:** Can amplify DNA from a single cell or a minute amount of biological material.
- **Speed:** PCR can generate results in a few hours, which is crucial in time-sensitive investigations.
- **Robustness:** PCR works even on **partially degraded samples**, allowing forensic scientists to work with less-than-ideal evidence.
- **Specificity:** Primers can be designed to target specific regions of DNA, making the analysis highly specific.

5. Challenges and Limitations:

- **Contamination:** PCR is highly sensitive, meaning that contamination from external DNA (e.g., from lab personnel) can result in inaccurate or misleading results.
- **Inhibition:** Substances like hemoglobin or chemicals found at crime scenes may inhibit the PCR reaction, leading to failure in amplification.

PCR's Role in Modern Forensic Science:

PCR revolutionized forensic science by making it possible to analyze DNA from small, old, or degraded samples that previously could not have been used. Its ability to amplify specific regions of DNA allows forensic scientists to create detailed DNA profiles that are crucial for solving crimes, identifying individuals, and ensuring justice is served.

DNA Profiling in Forensics

DNA profiling is a powerful forensic tool used to identify individuals based on their unique DNA sequences. It has revolutionized forensic science by allowing investigators to link biological evidence (e.g., blood, hair, or skin cells) to specific individuals, providing critical evidence in criminal investigations, paternity cases, and disaster victim identification.

1. What is DNA Profiling?

DNA profiling (also known as **genetic fingerprinting** or **DNA fingerprinting**) is the process of analyzing specific regions of an individual's DNA that are highly variable between people. These regions consist of **short tandem repeats (STRs)**, which are sections of non-coding DNA where sequences of bases are repeated multiple times. The number of repeats varies from person to person, making STRs ideal for identification.

2. Key Steps in DNA Profiling:

DNA profiling involves several steps to create a genetic fingerprint:

- **Collection of DNA Samples:**
 - DNA can be extracted from a wide variety of biological materials, including blood, saliva, hair, skin cells, semen, bone, and teeth.
 - Forensic scientists collect samples from crime scenes, victims, and suspects. Care is taken to avoid contamination of evidence.
- **DNA Extraction:**
 - DNA is extracted from the collected samples. In forensics, even small or degraded samples can yield enough DNA for analysis.
- **Amplification using PCR:**
 - Once DNA is extracted, it is often **amplified using PCR (Polymerase Chain Reaction)** to ensure there is enough material for analysis. PCR specifically targets STR regions.

- **STR Analysis:**

- The DNA profile is generated by analyzing **short tandem repeats (STRs)**. These are specific regions of the DNA where the same sequence of bases repeats multiple times.
- The number of repeats in these STR loci varies greatly between individuals, making them highly useful for distinguishing between people.
- In forensic DNA profiling, scientists typically analyze a set of **13 to 20 STR loci**. The probability of two unrelated people having the same number of repeats at all these loci is extremely low.

- **Gel Electrophoresis / Capillary Electrophoresis:**

- Once PCR amplifies the STR regions, the DNA fragments are separated based on their size using **gel electrophoresis** or **capillary electrophoresis**. This step allows scientists to determine the number of repeats at each STR locus.

- **Comparison of DNA Profiles:**

- The resulting **DNA profile** (a series of numbers that represent the number of repeats at each STR locus) is compared with the DNA profiles of known individuals, victims, or suspects.
- If two profiles match, it suggests that the samples came from the same person or are from biologically related individuals.

3. Uses of DNA Profiling in Forensics:

- **Criminal Investigations:**

- DNA found at crime scenes (e.g., blood, hair, or skin cells) can be matched to suspects, victims, or other individuals.
- DNA profiling has been instrumental in **identifying perpetrators** in crimes such as murder, sexual assault, and robbery. It can also help **exclude innocent individuals** from investigations.

- **Paternity and Kinship Testing:**

- DNA profiling is commonly used in paternity cases to establish biological relationships between a child and potential parents.
- It is also used in **family reunion cases** or to identify relatives of missing persons.

- **Identifying Remains (Disaster Victim Identification):**

- DNA profiling is vital in identifying victims of mass disasters, such as plane crashes or natural disasters, where bodies may be severely damaged or unrecognizable.
- DNA from relatives can be used to match with the remains of victims to confirm identities.

- **Exoneration of Innocent Individuals:**

- DNA profiling has helped to exonerate individuals who were wrongfully convicted. By reanalyzing old evidence using modern DNA techniques, forensic scientists can prove that a convicted individual was not present at the crime scene.

- **Cold Case Investigations:**

- DNA profiling has been used to solve **cold cases**, where evidence that was previously unusable or unprocessed has been re-examined, sometimes decades after the crime occurred.

4. DNA Databases:

Many countries have established **national DNA databases** where DNA profiles from convicted criminals, arrestees, and unsolved crime scene evidence are stored. These databases allow law enforcement to:

- **Search for matches** between crime scene DNA and individuals already in the database.
- **Link unsolved cases** by matching DNA evidence across different crime scenes (e.g., linking serial offenders).

In the UK, the **National DNA Database (NDNAD)** is one of the largest in the world, while in the US, the **Combined DNA Index System (CODIS)** serves a similar purpose.

5. Advantages of DNA Profiling:

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- **High specificity:** DNA profiling provides highly accurate identification because the likelihood of two unrelated individuals sharing the same STR profile is extremely low.
 - **Small sample sizes:** Even trace amounts of DNA (e.g., from a single hair or a small drop of blood) can be analyzed using PCR to produce a profile.
 - **Time-saving:** The process of creating a DNA profile is relatively fast and can be done in a matter of hours in some cases.
 - **Crime-solving:** DNA profiling is invaluable in solving both current and cold cases, bringing justice to victims and closure to families.

6. Limitations and Challenges:

- **Contamination:** DNA samples can be easily contaminated by DNA from investigators, other evidence, or lab personnel, leading to inaccurate results.
- **Partial profiles:** Degraded samples may yield only partial DNA profiles, making it harder to obtain a match or increasing the possibility of false exclusions.
- **Ethical concerns:** The creation and use of DNA databases raise privacy concerns about how genetic information is stored, accessed, and used by law enforcement.
- **Mixed samples:** In cases where DNA from multiple individuals is mixed (e.g., a crime scene with multiple people involved), it can be difficult to separate and analyze individual profiles.